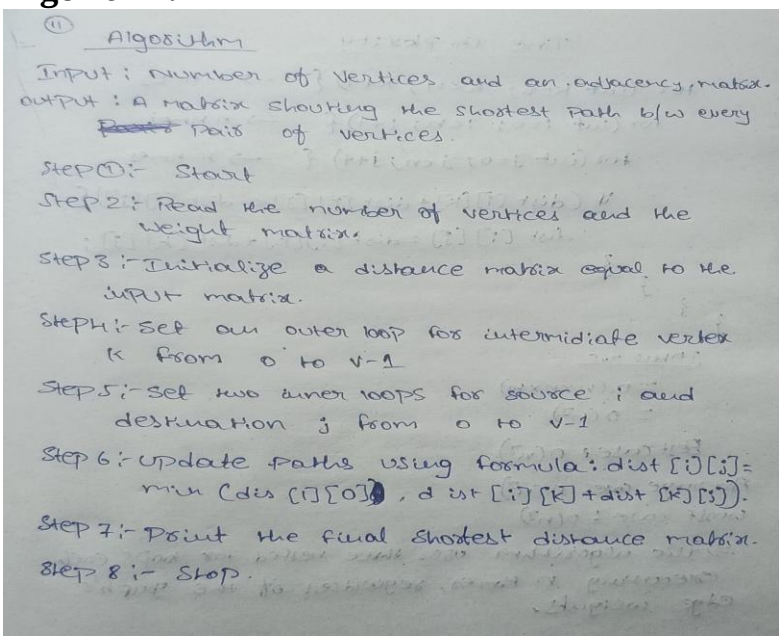
Practical –11

Aim: Write a program for Floyd-Warshal algorithm.

Algorithm:



(11) Algorithm

Input : number of vertices and an adjacency matrix.
output : A matrix showing the shortest path b/w every pair of vertices.

Step (1) :- Start
Step 2 :- Read the number of vertices and the weight matrix.
Step 3 :- Initialize a distance matrix equal to the input matrix.
Step 4 :- Set an outer loop for intermidiate vertex k from 0 to V-1
Step 5 :- Set two inner loops for source i and destination j from 0 to V-1
Step 6 :- Update paths using formula : dist[i][j] = min (dist[i][0], dist[i][k] + dist[k][j]).
Step 7 :- Print the final shortest distance matrix.
Step 8 :- Stop.

Program:

```
#include <iostream>

using namespace std;


int main() {

    int n;

    int dist[10][10];


    cout << "Enter number of vertices: ";
```

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```
cin >> n;

cout << "Enter the adjacency matrix:\n";
cout << "(Enter 99999 if there is no direct edge)\n";

for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        cin >> dist[i][j];
    }
}

// Floyd-Warshall Algorithm
for (int k = 0; k < n; k++) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (dist[i][k] != 99999 &&
                dist[k][j] != 99999) {

                if (dist[i][k] + dist[k][j] < dist[i][j]) {
                    dist[i][j] =
                        dist[i][k] + dist[k][j];
                }
            }
        }
```

```
        }
    }
    cout << "\nShortest Distance Matrix:\n";

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (dist[i][j] == 99999)
                cout << "INF\t";
            else
                cout << dist[i][j] << "\t";
        }
        cout << endl;
    }

    return 0;
}
```

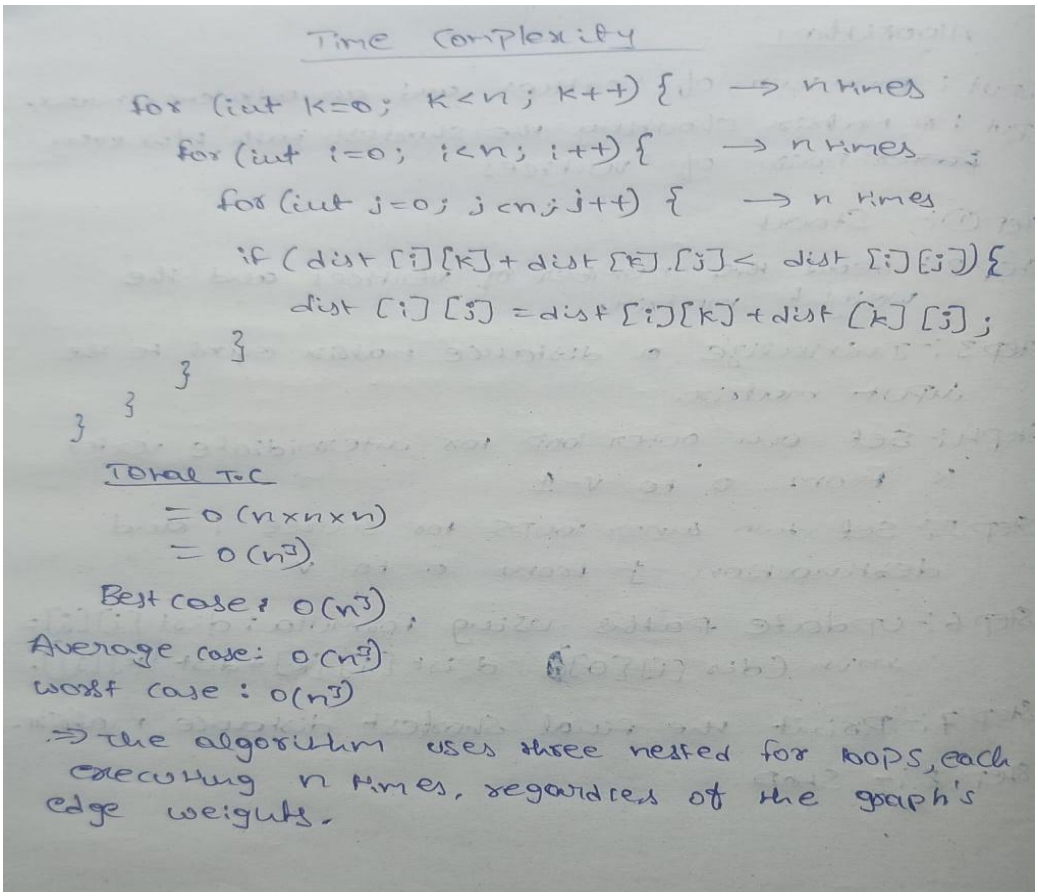
Output:

```
Enter number of vertices: 4
Enter the adjacency matrix:
(Enter 99999 if there is no direct edge)
0 5 99999 10
99999 0 3 99999
99999 99999 0 1
99999 99999 99999 0

Shortest Distance Matrix:
0   5   8   9
INF 0   3   4
INF INF 0   1
INF INF INF 0
```

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Time complexity :



Conclusion :

Thus, we have successfully implemented the Floyd–Warshall Algorithm in C++ to find the shortest paths between all pairs of vertices in a weighted graph. The algorithm updates the distance matrix by considering each vertex as an intermediate vertex. The best-case, average-case, and worst-case time complexities are all \(O(n^3)\).

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